ROOFTOP SOLAR PROJECT CRM

Business & Functional Requirement Document

Prepared by: NIRT Renewable Energy Private Limited

Purpose: Development of an End-to-End CRM for Rooftop Solar Projects

Document Status: Requirement Document for CRM Development

1. Introduction

NIRT Renewable Energy Private Limited is engaged in the rooftop solar business and handles projects from the initial customer enquiry up to project completion, net metering, subsidy claim and final closure.

At present, different activities are handled by different departments and persons. As the number of projects increases, it is important to have a **single integrated CRM system** to manage the complete project lifecycle.

The proposed CRM should provide a **single source of information** for every customer and every rooftop solar project.

The system should enable the management and employees to:

- Track every lead and customer.
- Track the status of every project.
- Assign responsibilities to employees.
- Set deadlines and follow-ups.
- Maintain all project documents.
- Track customer payments.
- Track DISCOM applications and approvals.
- Track procurement and material requirements.
- Track installation and commissioning.
- Track net metering.
- Track subsidy applications and receipts.
- Monitor pending activities.
- Generate MIS and management reports.
- Identify delays and bottlenecks.
- Maintain complete project history.

2. Main Objective of the CRM

The primary objective is to develop an **End-to-End Rooftop Solar Project Management CRM** covering the following complete process:

Lead Generation → Lead Qualification → Site Survey → Proposal → Customer Confirmation → Advance → Documentation → DISCOM Application → Technical Feasibility → Procurement → Installation → Inspection/Commissioning → Net Metering → Final Payment → Subsidy Claim → Subsidy Receipt → Project Closure

The CRM should ensure that **no lead, customer, activity, payment, document or project milestone is missed.**

3. Key Principles of the System

The CRM should be designed around the following principles:

3.1 One Customer – One Record

Every customer should have one master customer record.

The record should contain:

- Customer name
- Contact details
- Address
- Site address
- Consumer number
- DISCOM details
- Electricity connection details
- Project details
- Payment details
- Documents
- Communication history
- Project status
- Subsidy status

3.2 One Project – One Complete Digital File

Each rooftop solar project should have a unique **Project ID**.

All activities, documents, payments, approvals, materials, installation details, inspection details and subsidy information should be linked to this Project ID.

3.3 Clear Responsibility

Every activity should have:

- Responsible person
- Department
- Assigned date
- Target date
- Actual completion date
- Status
- Remarks

The system should clearly answer:

Who is responsible for this activity?

and

What is pending and why?

3.4 Complete Visibility

Management should be able to see the status of all projects from a single dashboard.

For example:

100 Projects

- 20 – Lead Stage
- 15 – Site Survey
- 10 – Proposal
- 12 – Customer Confirmation
- 8 – DISCOM Process
- 15 – Procurement
- 10 – Installation
- 5 – Net Metering
- 3 – Subsidy
- 2 – Closure

4. User Roles

The CRM should support role-based access.

Suggested users include:

1. Super Admin
2. Management
3. CRM Manager

4. CRM Executive
5. Sales Team
6. Site Survey Team
7. Design/Engineering Team
8. Purchase Team
9. Store Team
10. Project Manager
11. Installation Team
12. Accounts Team
13. Liaisoning/Documentation Team
14. Subsidy Team
15. Customer

Each user should see only the information and functions relevant to their role, based on management permissions.

5. Lead Management

5.1 Lead Creation

Leads may come from:

- Direct enquiry
- Website
- Phone call
- WhatsApp
- Referral
- Existing customer
- Exhibition/Expo
- Digital marketing
- Employee reference
- Channel partner
- Other sources

The CRM should allow easy lead creation.

Lead information should include:

- Lead ID
- Date

- Customer name
- Mobile number
- Email
- Location
- Lead source
- Interested system capacity
- Approximate monthly electricity bill
- Consumer category
- DISCOM
- Type of building
- Roof type
- Expected investment
- Subsidy eligibility
- Sales person
- Remarks

6. Lead Qualification

The CRM should provide a qualification process before the lead is converted into a project.

Possible status:

New → Contacted → Qualified → Site Survey Required → Not Qualified → Lost → Converted

The system should record the reason for:

- Lead rejected
- Customer not interested
- Price issue
- Technical issue
- Competitor selected
- Customer postponed
- Other reason

7. Site Survey Management

Once a lead is qualified, the site survey should be assigned to the concerned person.

The CRM should generate a **Site Survey Task**.

Site Survey information should include:

- Customer details
- Site address
- Survey date
- Surveyor
- Building type
- Roof type
- Roof area
- Available installation area
- Shadow analysis
- Existing electrical system
- Transformer details, if applicable
- Existing meter details
- Sanctioned load
- Connected load
- Average electricity consumption
- Recommended solar capacity
- Proposed module location
- Inverter location
- Cable route
- Earthing requirements
- Structural observations
- Photographs
- Videos
- Survey remarks

The surveyor should be able to upload photographs and documents directly from a mobile device.

8. Technical & Commercial Proposal

Based on the site survey, the CRM should support preparation and tracking of the customer proposal.

The proposal should include:

- Proposed system capacity
- Solar modules
- Inverter
- MMS/structure
- DC cables
- AC cables
- Earthing
- Protection system
- Installation charges
- Civil works
- Other components
- GST
- Total project value
- Subsidy, if applicable
- Customer payable amount
- Payment terms
- Expected generation
- Estimated savings
- Payback period

The CRM should maintain the complete history of proposals and revisions.

9. Customer Confirmation

Once the customer accepts the proposal, the CRM should convert the lead into a **Confirmed Project**.

The system should generate:

- Unique Project ID
- Customer ID
- Project registration date
- Confirmed capacity
- Project value
- Salesperson
- Project Manager

- Expected completion date

The project should then move to the next stage.

10. Advance Payment Management

The CRM should record customer advance payments.

Payment information:

- Project ID
- Invoice number
- Payment date
- Amount
- Payment mode
- Transaction/reference number
- Bank details
- GST details
- Balance amount
- Payment status

The system should automatically calculate:

Total Project Value – Amount Received = Balance Amount

Payment receipts should be uploadable and retrievable.

11. Customer Documentation

The CRM should have a document checklist based on project type.

Possible documents:

- Aadhaar
- PAN
- Electricity bill
- Consumer number
- Bank details
- Cancelled cheque
- Property documents, where required
- Customer photograph
- Application forms

- Declaration forms
- DISCOM documents
- Subsidy-related documents
- Other required documents

The system should clearly show:

Submitted / Pending / Not Applicable / Rejected

The CRM should automatically alert the responsible person when a required document is pending.

12. DISCOM Application Management

The CRM should maintain complete information relating to the DISCOM application.

Information should include:

- Application date
- Application number
- Consumer number
- DISCOM
- Division/Section
- Applied capacity
- Application status
- Application fee
- Payment details
- Documents submitted
- Approval date
- Rejection details
- Remarks

Possible status:

Not Started → Application Prepared → Submitted → Under Process → Approved → Rejected → Resubmission

13. Technical Feasibility

The CRM should provide a dedicated Technical Feasibility stage.

Information may include:

- Feasibility application date

- Feasibility status
- Approved capacity
- Technical conditions
- Additional requirements
- Transformer requirements
- Meter requirements
- Line requirements
- DISCOM observations
- Approval date
- Remarks

Any technical rejection or additional requirement should be recorded.

14. Procurement Management

Once the project is technically confirmed, procurement should be initiated.

The CRM should provide visibility of material requirements.

Procurement information:

- Project ID
- Material
- Specification
- Quantity
- Required date
- Purchase request date
- Purchase order date
- Vendor
- Expected delivery date
- Actual delivery date
- Quantity received
- Pending quantity
- Purchase status

Major materials may include:

- Solar modules
- Inverters

- MMS
- DC cables
- AC cables
- Earthing materials
- ACDB/DCDB
- Protection equipment
- Connectors
- Mounting accessories
- Civil materials
- Other project-specific materials

15. Store Management

The CRM should preferably be integrated with inventory/store management.

The system should track:

Required → Purchased → Received → Allocated → Issued → Installed

For each project, management should be able to see:

- Materials required
- Materials available
- Materials issued
- Materials pending
- Materials returned
- Material shortage

16. Installation Management

Once materials are available, installation should be assigned to the project team.

The installation module should track:

- Installation start date
- Installation completion date
- Installation team
- Module installation
- MMS installation
- Inverter installation

- DC cable installation
- AC cable installation
- Earthing
- Protection system
- Civil works
- Electrical works
- Testing
- Photographs
- Completion remarks

The Project Manager should be able to update the project progress.

17. Project Progress Tracking

The CRM should have a project progress tracker.

Example:

Stage	Status
Lead	Completed
Site Survey	Completed
Proposal	Completed
Customer Confirmation	Completed
Advance	Completed
Documentation	Completed
DISCOM Application	Completed
Technical Feasibility	Completed
Procurement	Completed
Installation	In Progress
Inspection	Pending
Net Metering	Pending
Final Payment	Pending
Subsidy	Pending
Project Closure	Pending

This should be visible on the project dashboard.

18. Inspection & Commissioning

The CRM should track inspection and commissioning.

Information should include:

- Inspection request date

- Inspection date
- Inspecting authority
- Inspection status
- Observations
- Rectification required
- Rectification completed
- Reinspection date
- Approval date
- Commissioning date
- Commissioning report
- Photographs
- Documents

Possible status:

Requested → Scheduled → Inspection Completed → Rectification → Approved → Commissioned

19. Net Metering

The CRM should have a separate Net Metering module.

Track:

- Application date
- Application number
- Meter details
- Net meter installation date
- Meter number
- Meter reading
- Status
- DISCOM communication
- Documents
- Completion date

Status:

Not Applied → Applied → Under Process → Approved → Meter Installed → Completed

20. Final Payment

The system should automatically calculate the outstanding customer amount.

Example:

Project Value: ₹2,50,000
Advance Received: ₹1,00,000
Other Payments: ₹50,000
Balance: ₹1,00,000

The CRM should generate payment reminders.

Payment status:

- Pending
- Partially Paid
- Fully Paid
- Overdue

The Accounts team should have a separate payment dashboard.

21. Subsidy Management

Where the customer is eligible for subsidy, the CRM should track the complete subsidy process.

Subsidy stages:

Eligibility → Application → Submitted → Under Verification → Approved → Subsidy Released → Subsidy Received

Information should include:

- Customer details
- Project ID
- Subsidy eligibility
- Eligible amount
- Application date
- Application number
- Documents
- Approval date
- Expected subsidy
- Actual subsidy received
- Receipt date
- Bank transaction/reference

- Remarks

22. Project Closure

A project should not be marked **Closed** until all mandatory activities are completed.

The system should have a Project Closure Checklist.

Closure checklist:

- Installation completed
- Testing completed
- Inspection completed
- Commissioning completed
- Net meter installed
- Customer documentation completed
- Final payment received
- Subsidy application completed, where applicable
- Subsidy received, where applicable
- Customer handover completed
- Warranty documents issued
- Completion certificate issued
- Project photographs uploaded
- Customer feedback obtained

Only after completion of mandatory requirements should the project be moved to:

PROJECT CLOSED

23. Customer Communication

The CRM should maintain a communication history.

Communication may include:

- Phone calls
- SMS
- Email
- WhatsApp
- Customer meetings
- Service requests

- Complaints
- Follow-ups

The system should maintain:

Date – Communication Type – Person – Subject – Remarks – Next Action

Where technically feasible, WhatsApp/email/SMS integration should be considered.

24. Task & Follow-up Management

Every activity should have a task.

Each task should contain:

- Task name
- Project ID
- Assigned person
- Assigned date
- Due date
- Priority
- Status
- Remarks
- Completion date

Priority:

High / Medium / Low

Status:

Pending / In Progress / Completed / Overdue / Cancelled

25. Alerts & Notifications

The CRM should generate automatic alerts for:

- Pending follow-ups
- Overdue activities
- Pending documents
- Pending payments
- DISCOM delays
- Procurement delays
- Installation delays

- Inspection dates
- Net metering pending
- Subsidy pending
- Customer complaints
- Project deadlines

Management should receive alerts for critical overdue activities.

26. Dashboard

The CRM should have a management dashboard.

Lead Dashboard

- Total leads
- New leads
- Qualified leads
- Lost leads
- Converted leads
- Lead conversion %
- Lead source performance

Sales Dashboard

- Proposals issued
- Proposals accepted
- Total project value
- Advance received
- Pending payments

Project Dashboard

- Total projects
- Projects in progress
- Projects delayed
- Projects completed
- Capacity under execution
- Capacity commissioned

Financial Dashboard

- Total project value

- Amount received
- Amount outstanding
- Overdue amount
- Subsidy receivable
- Subsidy received

Operations Dashboard

- Projects awaiting survey
- Projects awaiting design
- Projects awaiting procurement
- Projects awaiting installation
- Projects awaiting inspection

- Projects awaiting net metering

27. Management MIS Reports

The CRM should provide downloadable reports in Excel/PDF wherever required.

Important reports:

1. Lead Report
2. Lead Conversion Report
3. Sales Report
4. Customer Report
5. Site Survey Report
6. Proposal Report
7. Project Status Report
8. Project Delay Report
9. DISCOM Application Report
10. Technical Feasibility Report
11. Procurement Report
12. Material Pending Report
13. Installation Report
14. Inspection Report
15. Net Metering Report
16. Payment Outstanding Report

17. Subsidy Report
18. Project Closure Report
19. Employee Performance Report
20. Customer Complaint Report
21. Project-wise Profitability Report
22. Monthly Management MIS

28. Project Delay Monitoring

One of the important requirements is **delay identification**.

The CRM should compare:

Planned Date vs Actual Date

For every major stage.

For example:

Activity	Planned	Actual	Delay
Site Survey	05 Oct	05 Oct	0
Proposal	07 Oct	08 Oct	1 day
DISCOM Application	12 Oct	18 Oct	6 days
Procurement	15 Oct	20 Oct	5 days
Installation	25 Oct	Pending	5+ days

The system should highlight overdue activities.

29. Project Timeline

Each project should have a visual timeline showing:

Lead → Survey → Proposal → Confirmation → Advance → Documentation → DISCOM → Feasibility → Procurement → Installation → Inspection → Net Metering → Payment → Subsidy → Closure

This should allow management to understand the project status within a few seconds.

30. Document Management

All project-related documents should be stored digitally under the Project ID.

Documents should be categorised into:

- Customer Documents
- Technical Documents

- Commercial Documents
- DISCOM Documents
- Procurement Documents
- Installation Documents
- Inspection Documents
- Net Metering Documents
- Payment Documents
- Subsidy Documents
- Closure Documents

The system should provide version control wherever required.

31. Search Facility

The CRM should provide a powerful search facility.

Users should be able to search using:

- Customer name
- Mobile number
- Project ID
- Consumer number
- Application number
- Invoice number
- Location
- Project capacity
- Project status
- Salesperson
- Project Manager

32. Mobile Accessibility

Since site survey, installation and inspection activities happen at customer locations, the CRM should be mobile-friendly.

Field employees should be able to:

- View assigned projects
- Update tasks

- Upload photographs
- Upload documents
- Update survey information
- Update installation status
- Record customer discussions
- Update project progress
- Capture location/GPS where appropriate
- Submit reports

33. Role-Based Approval System

The CRM should support approval workflows.

For example:

Proposal → Approval → Customer Confirmation

Purchase Request → Approval → Purchase Order

Payment Request → Approval → Payment

Project Completion → Verification → Closure

Approvals should be traceable with:

- Approver
- Date
- Time
- Decision
- Remarks

34. Audit Trail

The CRM should maintain an audit trail.

The system should record:

- Who created the record
- Who modified the record
- What was modified
- Date
- Time
- Previous value

- New value

This is important for management control and accountability.

35. Customer Service / Complaint Management

The CRM should continue to support the customer even after project completion.

Customers should be able to raise:

- Service complaints
- Generation-related issues
- Inverter issues
- Module issues
- Net metering issues
- Warranty claims
- Maintenance requests

The system should track:

Complaint → Assignment → Action → Resolution → Customer Confirmation → Closure

36. Future AMC / O&M Module

The system should preferably be designed so that AMC/O&M can be added later.

This may include:

- AMC start date
- AMC expiry date
- Preventive maintenance schedule
- Service calls
- Breakdown calls
- Generation monitoring
- Inverter service
- Cleaning schedule
- Warranty tracking

37. Integrations

The CRM should be designed with future integration capability.

Possible integrations:

- WhatsApp
- Email
- SMS
- Payment gateway
- Accounting software
- ERP
- Zoho applications, where applicable
- DISCOM-related systems/API, where officially available
- Customer portal
- Mobile application

The development company should clearly specify which integrations are included in the current scope and which are future options.

38. Data Security

The CRM should provide adequate security for customer and company data.

Requirements should include:

- Secure login
- Role-based access
- Password policy
- Data backup
- Data recovery
- User activity logs
- Document access control
- Secure cloud hosting
- Data encryption where appropriate
- Protection against unauthorised access

39. Master Data Management

The system should have master data for:

- Customers
- Employees

- Departments
- DISCOMs
- Locations
- Vendors
- Materials
- Solar modules
- Inverters
- Project types
- Payment types
- Lead sources
- Project stages
- Complaint categories
- Document types

Master data should be editable only by authorised users.

40. Workflow Configuration

The CRM should preferably allow authorised administrators to configure:

- Project stages
- Status values
- Task types
- Approval levels
- Document checklists
- Notification rules
- User roles
- Escalation periods
- Mandatory fields

This will reduce the need for software modification whenever the business process changes.

41. Important Business Rule

The CRM should not merely act as a **customer database**.

It should function as a **Project Control System**.

For every project, management should be able to answer the following questions:

1. Who is the customer?
2. What is the project capacity?
3. What is the project value?
4. How much money has been received?
5. How much is pending?
6. What stage is the project currently in?
7. Who is responsible for the current stage?
8. What is pending?
9. How many days has it been pending?
10. Why is it pending?
11. What is the next action?
12. Who should take the action?
13. When should it be completed?
14. What documents are pending?
15. What materials are pending?
16. What approvals are pending?
17. Is the project delayed?
18. What is the expected completion date?
19. Is subsidy applicable?
20. Has the subsidy been received?
21. Can the project be closed?

42. Overall Project Status

The CRM should use a standard project status structure.

Suggested Status

1. Lead

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2. Qualified

↓

3. Site Survey

↓

4. Proposal

↓

5. Customer Confirmation

↓

6. Advance Received

↓

7. Documentation

↓

8. DISCOM Application

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9. Technical Feasibility

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10. Procurement

↓

11. Installation

↓

12. Inspection / Commissioning

↓

13. Net Metering

↓

14. Final Payment

↓

15. Subsidy Claim

↓

16. Subsidy Receipt

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17. Project Closure

43. CRM Home Page – Recommended View

When management logs into the CRM, the home page should immediately show:

TODAY

- New Leads
- Follow-ups Due
- Site Surveys Today
- Customer Meetings
- Payments Due
- Projects Due for Action

- Overdue Tasks

PROJECTS

- Total Projects
- Projects in Progress
- Projects Delayed
- Projects Completed
- MW/kW Under Execution
- MW/kW Commissioned

FINANCE

- Total Order Value
- Amount Received
- Outstanding Amount
- Overdue Amount
- Subsidy Receivable
- Subsidy Received

OPERATIONS

- Procurement Pending
- Installation Pending
- Inspection Pending
- Net Metering Pending
- Documentation Pending

44. Expected Output from the CRM Development Company

The software development company is expected to provide:

1. Business requirement analysis
2. System architecture
3. UI/UX design
4. Web-based CRM
5. Mobile-responsive interface
6. User management
7. Role-based access
8. Complete workflow development

9. Database design
10. Document management
11. Dashboard
12. MIS reports
13. Notifications and alerts
14. Approval workflow
15. Audit trail
16. Data backup
17. Security
18. Testing
19. User acceptance testing
20. Training
21. Data migration, if required
22. Deployment
23. Documentation
24. Post-implementation support
25. Maintenance and AMC proposal

45. Development Approach Expected

Before starting full development, the CRM company should first conduct detailed discussions with NIRT's key users.

The suggested approach is:

Requirement Discussion

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Process Mapping

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BRD Finalisation

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UI/UX Prototype

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Management Approval

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Development

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Module-wise Testing

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User Acceptance Testing (UAT)

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Corrections

↓

Final Deployment

↓

User Training

↓

Go-Live

↓

Post-Go-Live Support

46. User Acceptance Testing

Before final acceptance, NIRT should be able to test the system using actual sample projects.

The vendor should demonstrate at least one complete project journey:

Lead → Survey → Proposal → Confirmation → Advance → Documents → DISCOM → Feasibility → Procurement → Installation → Inspection → Net Metering → Payment → Subsidy → Closure

The project should be traceable from beginning to end.

47. Important Expectation from the Developer

The development company should not develop the system based only on the process flow provided in this document.

They are expected to:

- Study NIRT's actual business process.
- Identify gaps.
- Suggest improvements.
- Identify duplicate activities.

- Recommend automation opportunities.
- Identify required approvals.
- Recommend useful dashboards.
- Identify integration opportunities.
- Design the system for future scalability.

Any deviation from the proposed workflow should be discussed with NIRT before implementation.

48. Scalability

The CRM should not be designed only for the current number of projects.

It should be capable of handling increasing:

- Customers
- Leads
- Projects
- Employees
- Documents
- Transactions
- Locations
- Project capacities

The architecture should allow additional modules to be added in the future.

Possible future modules include:

- O&M
- AMC
- Customer Portal
- Mobile App
- Inventory
- Vendor Management
- Accounting Integration
- HR Integration
- Project Costing
- Profitability Analysis
- Generation Monitoring

- Service Management

49. Key Success Criteria

The CRM will be considered successful if management can obtain the following information **within a few clicks**:

- How many leads do we have?
- How many have been converted?
- What is the total business value?
- Which projects are currently active?
- Which projects are delayed?
- Who is responsible for the delay?
- What payments are pending?
- Which documents are pending?
- Which DISCOM applications are pending?
- Which projects are waiting for procurement?
- Which projects are waiting for installation?
- Which projects are waiting for inspection?
- Which projects are waiting for net metering?
- Which customers are eligible for subsidy?
- How much subsidy is pending?
- Which projects are ready for closure?

50. Conclusion

NIRT Renewable Energy Private Limited intends to develop an **integrated, user-friendly and scalable Rooftop Solar CRM** that provides complete visibility and control over the customer and project lifecycle.

The CRM should reduce manual follow-up, improve accountability, minimise delays, improve customer communication, strengthen documentation and provide management with accurate real-time information.

The ultimate objective is:

"One Customer – One Project – One Digital Record – Complete Visibility – Clear Accountability."

The CRM should therefore be designed not merely as a **Lead Management System**, but as a complete:

Rooftop Solar Project Lifecycle Management System

covering the journey from:

Lead Generation → Customer Acquisition → Project Execution → Payment → Subsidy → Project Closure

with complete **people, process, document, financial and management control**.